

except 4, 6 and 8 while Q_4 consists of images of all of them except 6 and 8. One possible explanation could be that these tests distinguish $\mathcal{P}$ from Q_3 mainly by detecting if images of the digit 6 appear in the sample, this would explain why we observe similar power for Q_3 and Q_4 , and why the power for Q_5 (consisting of every digit except 8) drops significantly.

We also sometimes observe in Figure 5 that our aggregated tests with different weighting strategies obtain slightly higher power for Q_4 than for Q_3 , which might at first seem counter-intuitive. This could be explained by the fact that the optimal bandwidths for distinguishing $\mathcal{P}$ from Q_3 and from Q_4 might be very different, and that the choice of collections of bandwidths presented in Figure 5 are slightly better suited for distinguishing $\mathcal{P}$ from Q_4 than from Q_3 . While it is also the case in Figures 3 and 4 that the alternatives with different number of perturbations require different bandwidths to be detected, it looks like in that case considering a collection of five bandwidths which are powers of 2 is enough to adapt to those differences. For the MNIST experiment in Figure 5, it seems that the differences between the optimal bandwidths for Q_3 and Q_4 are more important. Using **MMDAgg*** with its adaptive parameter-free collection of bandwidths solves this problem. An advantage of our aggregated MMDAgg tests is that, even if we fix the collection of bandwidths, they are able to detect differences at various lengthscales, this is not the case for the **median** and **split** tests as those select some specific bandwidth and are only able to detect the differences at the corresponding lengthscale.

5.7 Power experiment: continuous limit of the collection of bandwidths

Our collection of bandwidths for **MMDAgg*** is a discretisation of an interval using $N = 10$ points, which we formally introduced in Section 5.2. As we increase the number of points

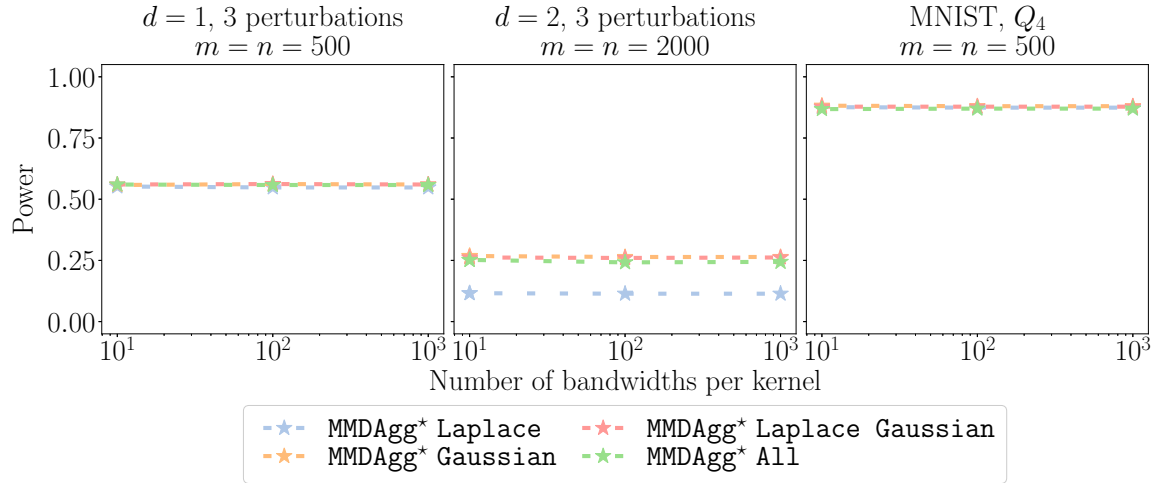


Figure 6: Power experiments varying the size of the collections of bandwidths using perturbed uniform d -dimensional distributions and the MNIST dataset with a wild bootstrap. For **MMDAgg* Laplace Gaussian**, two kernels are aggregated, each with a varying number of bandwidths. For **MMDAgg* All**, 12 kernels are considered with a varying number of bandwidths (up to 12000 kernels are aggregated).

N , the discretisation becomes finer and in the limit as $N \rightarrow \infty$ it corresponds to the whole continuous interval. In Figure 6, we consider the three experiments of Figures 3 to 5 presented in Sections 5.5 and 5.6 for **MMDAgg* Gaussian**, **MMDAgg* Laplace**, **MMDAgg* Laplace Gaussian**, and **MMDAgg* All**, with collections of sizes N , N , $2N$ and $12N$, respectively (details in Section 5.4). We vary the number of bandwidths per kernel N to be 10, 100 and 1000. In particular, this means for example that **MMDAgg* All** with $N = 1000$ aggregates over $12N = 12000$ kernels. For the one-dimensional uniform setting, we use three perturbations and sample sizes 500. The 2-dimensional case is considered with three perturbations and $m = n = 2000$. For the MNIST experiment, we use Q_4 as an alternative (every digit except 8 and 6) with sample sizes 500.

As in the previous experiments, in Figure 6, the MMDAgg test aggregating both Laplace and Gaussian kernels obtains the highest power achieved by either **MMDAgg* Laplace** or **MMDAgg* Gaussian**. Considering more types of kernels with **MMDAgg* All** does not change the test power. Since it is computationally more expensive, we recommend using **MMDAgg* Laplace Gaussian** in practice.

In Figure 6, we observe for all four tests that the test power remains the same when increasing the number of bandwidths per kernel from 10 to 1000. The case $N = 1000$ simulates the continuous limit of the collection of bandwidths, that is, when the full interval is considered without discretisation. First, this shows that our MMDAgg test retains high power even when aggregating up to 12000 kernels, which is only possible due to the way we perform the level correction. Second, this illustrates that the power of the aggregated test with a continuous collection of bandwidths, can also be achieved by the less computationally expensive MMDAgg test with a discretisation of $N = 10$ points.

5.8 Power experiment for image shift detection

In this section, we consider the experiment of Rabanser et al. (2019, Table 1a) on image shift detection on the MNIST (LeCun et al., 2010) and CIFAR-10 (Krizhevsky, 2009) datasets. Ten different types of shifts are applied to either 10%, 50% or 100% of the samples, those include adversarial shifts, class knock-outs, injecting Gaussian noise, and combining rotations/translations/zoom-ins, with different shift strengths (see Section 4 of Rabanser et al. 2019 for details).

We report in Table 1 the test power obtained by our four **MMDAgg*** tests, by our **MMDAgg uniform** test with Laplace kernels and with Gaussian kernels, as well as by four versions of the **AutoML** test of Kübler et al. (2022b, Table 1a). We observe that the **MMDAgg*** tests outperform the **MMDAgg uniform** tests. For **MMDAgg***, using the Laplace kernel performs the best, aggregating both Laplace and Gaussian kernels performs better than using only Gaussian kernels and almost as well as using only Laplace kernels, aggregating many more kernels with **MMDAgg* All** results in the same power obtained by **MMDAgg* Laplace Gaussian**.

Despite using off-the-shelf kernels which are not specifically designed for images, our aggregated **MMDAgg*** tests are still performing within only a few percentage points of state-of-the-art tests based on training models (*e.g.* neural networks) which excel on image data, such as the **AutoML** test.

Table 1: Image shift detection experiment of Rabanser et al. (2019). The numbers reported correspond to the test power averaged over 60 alternatives (each repeated 5 times): 10 different shift types applied on either 10%, 50% or 100% of the image samples drawn from either the MNIST or CIFAR-10 datasets.

Test	Number of samples							
	10	20	50	100	200	500	1000	10000
MMDAgg* Laplace	0.21	0.29	0.40	0.44	0.47	0.56	0.67	0.83
MMDAgg* Gaussian	0.19	0.26	0.34	0.42	0.42	0.51	0.62	0.75
MMDAgg* Laplace Gaussian	0.21	0.27	0.37	0.43	0.45	0.54	0.65	0.80
MMDAgg* All	0.21	0.27	0.37	0.43	0.46	0.55	0.66	0.80
MMDAgg uniform (Laplace)	0.20	0.28	0.40	0.43	0.46	0.52	0.58	0.79
MMDAgg uniform (Gaussian)	0.15	0.23	0.33	0.35	0.38	0.44	0.48	0.69
AutoML (raw)	0.17	0.24	0.37	0.46	0.50	0.62	0.67	0.87
AutoML (pre)	0.18	0.29	0.42	0.47	0.47	0.64	0.65	0.72
AutoML (class)	0.19	0.19	0.38	0.46	0.52	0.61	0.67	0.87
AutoML (bin)	0.03	0.14	0.31	0.43	0.49	0.51	0.59	0.86

5.9 Overview of additional experimental results

We consider additional experiments in Appendix A, we briefly summarize them here. In Appendix A.1, we verify that all the tests we consider have well-calibrated levels. We then consider widening the collection the bandwidths for our tests **MMDAgg uniform** and **MMDAgg centred** in Appendix A.2, and observe that the associated cost in the power is relatively small. We verify in Appendix A.3 that using a wild bootstrap or permutations results in similar performance; the difference is of non-significant order and is not biased towards one or the other. In Appendix A.4, we show that if one of the sample sizes is fixed to a small number, we cannot obtain high power even if we take the other sample size to be very large. Finally, we increase the sample sizes for the **ost** test in Appendix A.5 and observe that we need extremely large sample sizes to match the performance of **MMDAgg uniform** with 500 or 2000 samples.

6 Conclusion and future work

We have constructed a two-sample hypothesis test, called **MMDAgg**, which aggregates multiple MMD tests using different kernels/bandwidths. Our test is adaptive over Sobolev balls and does not require data splitting. We have proved that **MMDAgg** is optimal in the minimax sense over Sobolev balls up to an iterated logarithmic term, for any product of one-dimensional translation invariant characteristic kernels which are absolutely and square integrable. This optimality result also holds under two popular strategies used in estimating the test thresholds, namely the wild bootstrap and permutation procedures. In practice, we propose four weighting strategies which allow the user to incorporate prior knowledge about the collection of bandwidths. We also introduce a parameter-free adaptive collection